Persistence Mechanics: Architecture, Dynamics, and the Operational Action

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Before a complex system can adapt or self-organize, it must remain identifiable through time. We formalize the mechanics of this persistence, deriving the substrate-independent, minimal architecture. We show that any persistent system must satisfy three fundamental constraints: Finiteness (bounded resources), Conservation (preserved information), and Causality (well-defined evolution), yielding six irreducible pillars: Capacity, Map, Protocol, Governor, Toll, and Margin. This static architecture is grounded in time by defining the minimal dynamical description of state transitions: toll (c), structural complexity (K), and coordinated transition flux (f) whose product defines the operational cost rate density $\mathcal{D} = c \cdot K \cdot f$. Integrating $\mathcal{D}$ over a system's history defines the Operational Action S_Φ. The resulting Selection Principle shows that minimum-action configurations asymptotically dominate observable histories. Mapping systems onto this architectural invariant provides a common basis for comparing maintenance constraints, identifying structurally analogous failure modes, and formulating cross-domain hypotheses for subsequent validation.

I. INTRODUCTION

From biological organisms to engineered feedback loops, a central problem in complex-systems science is how systems retain recognizable organization despite disturbance, turnover, and decay [1]. Several mature disciplines address particular problems faced by such systems: control theory characterizes regulation and stability; information theory characterizes representation and communication under noise; and thermodynamics constrains energy conversion and dissipation. Each begins with a system boundary, state space, and relevant mechanisms already specified. This leaves a prior cross-domain question: what minimal architecture must any system instantiate simply to remain identifiable through time?

Previous complex-systems research has explored complementary aspects of persistence. Prigogine characterized minimum entropy production in stationary states under linear near-equilibrium conditions [2]; Friston related adaptive self-maintenance to variational free-energy minimization through an internal model [3]; England examined how external driving can favor particular configurations in nonequilibrium systems [4]; and cybernetics established organizational constraints on effective regulation, including requisite variety and the Good-Regulator theorem [5, 6]. These approaches characterize complementary aspects of persistence. Comparing them across domains requires a common criterion for what it means for a system to persist.

We say that a system *persists* across an observation interval if, relative to a defined boundary and class of disturbances, it maintains the organization that defines it and stays within its viable operating limits. Persistence does not demand static components or fixed internal states. Rather, components may turn over and resources may flow through the system, provided its defining organization remains intact.

Persistence Mechanics studies the general conditions governing systems that persist: the requirements, constraints, and behaviors shared across substrates. Here we addresses three questions: what must be instantiated for persistence, how the resulting architecture operates through time, and how its cumulative costs affect which histories remain observable. Section II derives the **Architecture of Persistence**: six substrate-independent pillars derived from the logical requirements of Finiteness, Conservation, and Causality. Section III develops the **Dynamics of Persistence**: identifying the transition toll c, structural complexity $\mathbf{K}$, and coordinated transition flux f. Section IV derives the exponential weighting of histories by cumulative Operational Action and states the conditions under which surviving histories concentrate on minimum-action trajectories. Section V extends the additive baseline to systems with external flux and shared-resource contention. Section VI illustrates how the six pillars are instantiated across biological, ecological, and computational substrates.

II. THE ARCHITECTURE OF PERSISTENCE

Before a system can grow or adapt, it must preserve its identity and survive. We organize the conditions for persistence into three core constraints: Finiteness, Conservation, and Causality. Because each requirement needs a way to be established and a way to be protected from disruption, this yields six functional pillars of persistence. This section defines those pillars, examines the failure modes they address, and shows how familiar capabilities can be composed from them.

A. The Structural Constraints

Any system persisting through time is bounded in extent, carries distinguishable content, and evolves through an ordered sequence of states. Absent any one, it is in-

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distinguishable from its environment, from noise, or from something static. Formally, these correspond to: **Finiteness** (bounded resources against divergence), **Conservation** (preserved information against dissipation), and **Causality** (well-defined temporal evolution against ambiguity).

To persist, each constraint must survive its characteristic perturbation. Each therefore demands not only **provision** (establishing the means to satisfy the constraint), but also **defense** (protecting that means from the characteristic failure mode) yielding the six functional pillars.

Constraint	Provision	Defense
Finiteness	Capacity	Margin
Conservation	Map	Protocol
Causality	Toll	Governor

Table I. The three constraints, organized by provision and defense, define the six pillars of the PM architecture.

A *pillar* is a functionally distinct condition that must be materially or informationally instantiated for persistence. Pillars need not correspond one-to-one with components: a single mechanism may instantiate several pillars, and one pillar may be distributed across many mechanisms.

These pillars operate upon a medium that imposes limits (bandwidth, latency, noise floor) referred to as a **Substrate**.

B. The Six Pillars

For each pillar, we define it and connect it to established mathematical results that provide a basis for formal analysis.

Capacity (*CAP*): The resources and infrastructure available to maintain the system's organization and carry out its operations. Capacity sets how much the system can store, process, and sustain. *Formal Connection (The Bekenstein Bound[7]):* For physical systems within its scope, Bekenstein's bound limits information storage in terms of energy and spatial extent. It provides a concrete realization of Capacity: the information a system can retain is constrained by its physical resources.

Margin (*MAR*): The reserves and operating tolerances that allow the system to absorb disturbances without leaving its viable range. *Formal Connection (Nyquist-Bode Stability Margins[8] and Kingman's Formula[9]):* In linear control, gain and phase margins quantify how much loop gain or phase can change before stability is lost. Similarly, Kingman's heavy-traffic approximation shows that, with nonzero arrival or service-time variability, mean waiting time grows without bound as queueing utilization approaches unity. Both results illustrate the role of operating margins in sustaining performance under uncertainty.

Map (*MAP*): The retained organization or relational structure that defines the system and preserves distinctions relevant to its continued operation. In adaptive systems, Map may be instantiated as an internal predictive model; in passive systems, it may be embodied directly in physical organization. *Formal Connection (The Good-Regulator Theorem [6]):* Conant and Ashby's Good-Regulator theorem establishes that, under its assumptions, the simplest optimal regulator must embody a model of the system it regulates.

Protocol (*PRO*): The mechanisms through which components exchange information and coordinate state transitions. Protocol supports internal coherence by preserving relevant distinctions across the system's interfaces. *Formal Connection (Shannon's Source-Channel Separation and Channel Capacity [10]):* Shannon's coding results characterize conditions for reliable communication, providing a realization of the Protocol.

Toll (*TOL*): The resource expenditure and time required to realize state transitions. *Formal Connection (Landauer's Principle and the Margolus–Levitin Limit[11, 12]):* Under standard isothermal erasure conditions, resetting one unbiased bit requires heat dissipation of at least $k_B T \ln 2$. The Margolus–Levitin limit bounds the time required for a quantum system to evolve to an orthogonal state at a specified mean energy above its ground state. These results formalize energetic and temporal constraints on state transitions.

Governor (*GOV*): The active or passive constraints that restrict which transitions the system can undergo, suppressing, redirecting, or terminating those that threaten its viable operation. *Formal Connection (Ashby's Law of Requisite Variety[5]):* Ashby's law bounds the regulatory variety required to restrict the effects of disturbances to an acceptable range of outcomes.

C. Necessity: Counterfactual Failure Modes

The necessity of the six pillars follows by analyzing the **Counterfactual Failure Mode** of a system lacking exactly one pillar. In every case, the system's viability collapses, limiting its expected lifetime τ to the onset of its characteristic perturbation.

- **No Capacity (Starvation):** Without baseline operational infrastructure, the system cannot acquire, process, or export resources. Once existing reserves are spent, any perturbation is fatal; $\tau \to 0$ via rapid exhaustion.
- **No Margin (Fragility):** Operating at theoretical criticality leaves zero tolerance for environmental variance. Any non-zero perturbation exceeds the

system's structural limits immediately, so the expected time to a fatal shock is finite; $\tau \to 0$ by random shock.

- **No Map (Dissolution):** Without an internal model, the system has no reference for coherence, and no boundary to maintain against its environment; $\tau \to 0$ as the system–environment boundary dissolves.
- **No Protocol (Decoherence):** Components act on statistically independent, outdated states, with no mechanism to reconcile them; $\tau \to 0$ by organizational fragmentation.
- **No Toll (Zeno Collapse):** A causal sequence requires irreducibly discrete state transitions. Without an irreducible toll per operation, the sequence can be infinitely subdivided; no individual step remains distinguishable as a cause of the next, and the ordering required for well-defined evolution dissolves. $\tau \to 0$ by causal decoherence.
- **No Governor (Divergence):** Unconstrained positive feedback drives resource consumption toward unbounded growth, exhausting the substrate or diverging past any recoverable bound; $\tau \to 0$ by unbounded runaway.

Since the removal of any component results in $\tau \to 0$, every pillar is necessary.

D. Completeness: Compositional Closure

We conjecture that the six pillars are sufficient to account for the functional requirements of persistence: no additional independent role is needed. We do not yet have a general proof. Every candidate requirement we have examined can be expressed as a composition of the six pillars, and we have identified no counterexample to date.

The following examples illustrate these compositions without enumerating every supporting role. Memory is Map content provisioned against temporal variance by Margin. Prediction is Map operation on incoming signal via Protocol. Repair is Governor-triggered reprovisioning within the Margin reserve. Replication is Protocol-mediated write of Map onto fresh Capacity. Feedforward networks, state observers, and adaptive controllers can likewise be described through combinations of the six roles, without requiring an additional primitive in these descriptions.

E. The Cost of Persistence

A system consumes resources by operating the six pillars. Expressed in common units, these costs provide a basis for comparing systems and, when related to available resources and failure mechanisms, their lifetimes.

To distinguish baseline structural costs from task-related expenditure, we evaluate baseline structural costs in a *quiescent steady state* (QSS), taken here as stationary operation at zero task flux. Internal housekeeping and supporting resource exchanges may continue.

Protocol and Governor incur operating costs but may reduce expenditure elsewhere. Relative to a specified uncoordinated baseline, their net contribution can be negative when the savings exceed their operating costs. Let Z_i denote the cost contribution allocated to pillar i:

$$\bar{Z}_{\text{net}} = Z_{\text{CAP}} + Z_{\text{MAP}} + Z_{\text{TOL}} + Z_{\text{MAR}} + Z_{\text{PRO}} + Z_{\text{GOV}} \quad (1)$$

Because the pillars occupy orthogonal domains at zero flux, any correlation between them is already priced into the individual coordination terms Z_{PRO} and Z_{GOV}; no additional cross-term is needed to capture coupling between pillars within this baseline. Coupling outside QSS introduces non-linear contention, which we formalize in section V on open systems.

F. Relation to Existing Disciplines

Existing disciplines encounter the six functional pillars in the systems they study. While these disciplines also investigate architecture, they primarily address domain-specific problems. PM's contribution is identifying the common underlying organization, regardless of substrate and domain. This gives comparisons across disciplines a shared framework whose implications are explored in section VI.

1. Relation to Information Theory

Information theory quantifies how representation, encoding, and noise constrain communication across physical mediums. In doing so, it relies on structural primitives that map directly to the six pillars: channel capacity (Capacity), codebook topologies (Map), framing and handshaking routines (Protocol), power constraints (Governor), processing overhead (Toll), and link margins (Margin). These problems are analyzed under specified source, channel, and resource assumptions.

2. Relation to Thermodynamics

Thermodynamics describes how physical systems evolve under energy exchange and thermal fluctuations. It relies on physical primitives that correspond to the six pillars: extensive state variables and phase space volume (Capacity), metastable energy landscapes (Map), kinetic transition pathways (Protocol), conservation laws (Governor), irreversible entropy production (Toll), and free-

energy activation barriers (Margin). These correspondences draw on thermodynamics, statistical mechanics, and kinetics, each with its own assumptions about states, exchanges, and transition mechanisms.

3. Relation to Control Theory

Classical control theory emerged from engineering physical servomechanisms: radar mounts, autopilots, and chemical plants whose historical context was optimizing systems already built, powered, and staffed. Its four given components (the plant, the sensor, the controller, and the actuator) are convergent evidence for four of the six pillars: Plant/Actuator (Capacity), Reference Signal/Setpoint (Map), Sensor/Feedback Path (Protocol), Controller/Limiter (Governor). The remaining two pillars are rarely canonized alongside the first four as they are not physical components, but describe how the four components behave under time and stress. Time delays, bandwidth limits, and processing latency (Toll) were minimized as engineering overhead and gain/phase margins (Margin) were tuned for stability. Control theory asks how to regulate an existing system and the classical four-component loop is a physical instantiation of the six functional roles.

III. THE DYNAMICS OF PERSISTENCE

To describe how a system maintains its architecture through time, we identify the minimal accounting of its state transitions: the work performed per update, the cost of that work, and the frequency of updates. These quantities define a local operational cost rate density. The finite operational budget presented in most persistent systems results in a trade-off between the three with unexpected implications.

A. State Transition Costs

The operation of the six pillars through time introduces the constraint that transitions must be realized within the substrate's available resources and response times. Quantifying this requires distinguishing the work an update requires, the cost of that work, and the update frequency.

1. **Structural workload (K):** The number of elementary operations required for one maintenance update of the local structure. An update preserves, repairs, or modifies the organization through which the pillars are instantiated. Here, K measures operations per update. Its relationship to algorithmic description length [13] is implementation-specific. *Domain Manifestation (Computational Workload):* In a computational implementation, K counts the elementary operations executed by an update procedure under a specified computational model.

2. **Operational toll (c):** The mean resource cost per elementary operation, expressed in a specified accounting unit. Substrate manifestations include dissipated energy, computational expenditure, or other explicitly defined resources. Transition latency imposes a separate constraint on achievable update rates; it is not automatically equivalent to dissipative cost. *Domain Manifestation (Information Erasure):* In the standard isothermal setting, erasing one unbiased bit requires at least $k_B T \ln 2$ of heat dissipation [11]. This provides a physical lower bound on the toll for that class of operation.

3. **Coordinated transition flux (f):** The number of maintenance updates performed per unit substrate measure per unit time. An update involves coordinated operations, whose total is recorded by K. The flux required for persistence is dictated by the system's organization and the environmental disturbances it must accommodate. *Domain Manifestation (Communication-Constrained Control):* Data-rate theorems establish communication requirements for stabilizing specified dynamical systems [14]. For updates carrying a fixed amount of transmitted information, such requirements can constrain update frequency.

While a system might be made up of sub systems each performing transitions, the parent system transition costs will only be the sum of the subsystem costs.

B. Operational Cost Rate Density

Repeated operations draw on a finite resource budget. Comparing this demand with available resources requires combining the workload, operational toll, and update flux into a local cost rate per unit substrate measure.

$$[c] = \frac{\text{cost}}{\text{operation}}$$

$$[K] = \frac{\text{operations}}{\text{update}}$$

$$[f] = \frac{\text{updates}}{\text{substrate measure} \cdot \text{time}}$$

For transitions requiring K elementary operations each, if the mean cost of those operations is c, the cost per update is Kc. Performing updates at flux f therefore yields the local density:

$$\mathcal{D} = cKf, \qquad [\mathcal{D}] = \frac{\text{cost}}{\text{substrate measure} \cdot \text{time}} \tag{2}$$

The interpretation of $\mathcal{D}$ depends on the resource used to define c. When c measures dissipated energy per operation, $\mathcal{D}$ is a dissipation density; otherwise it records the corresponding operational resource cost.

When any operational cost is 0, $\mathcal{D} = 0$. Whether the corresponding inactivity or absence of expenditure compromises persistence depends on the maintenance required over the observation interval.

C. The Complexity Trade-off

Transitions draw on a resource budget whose sustainable rate $\mathcal{D}_{\max}$ is set by the substrate and environment. This budget constrains the combinations of workload, toll, and transition flux that the system can sustain and results in a competition among the three factors.

Any persistent architecture is constrained to the feasible operating surface $cKf \leq D_{max}$. At a fixed budget, increasing workload requires a compensating reduction in toll, transition flux, or both.

For illustration, consider a cell with a fixed ATP budget. If each update requires twice as many ATP-consuming operations while the ATP cost per operation remains unchanged, the same budget supports only half as many updates per unit time. Maintaining the original update rate would require either a lower cost per operation or a larger budget.

Because this applies to all persistent systems various properties that might be unintuitive can be assumed. When budgets are fixed knowing two of the values in a system will let one accurately predict the third and when one value changes one or both of the other values are guaranteed to change. Conversely if the budget changes; one, two or all three values will change.

D. Operational Clocks

State transitions provide an operational basis for counting time when successive updates serve as clock ticks.

a. Budget Because each tick requires resources the operational budget bounds the maximum tick rate. Conversely any system that persists against external disturbance might have a low tick rate, but never zero.

b. Relative Clocks Relative to a common external time reference, systems with different clock rates accumulate different tick counts over the same interval.

c. Necessary responses When persistence requires a response to an external threat within a deadline, the required updates impose a minimum resource allocation that must be dedicated to f to respond in time. If the clock and the operations it measures slow proportionally, their internal timing relationships remain unchanged, even as the update rate potentially falls below that required to meet the deadline without detection through internal comparisons.

IV. MINIMUM ACTION AS SURVIVORSHIP

The Operational Action defines what a system must minimize, but not which histories survive to be observed. The selection filter derives from the multiplicative compounding of independent transition risk, and show that only minimal-action configurations remain observable.

A. Operational Action

While the operational cost rate describes instantaneous expenditure, evaluating the accumulated burden of distinct histories requires integrating this rate across the system's spatial extent and temporal duration.

Let $\Sigma(t)$ denote the physical or computational elements defining the system boundary at time t. Integrating the local cost-rate density over this footprint yields the total instantaneous operational cost:

$$Z_{\text{net}}(t) = \int_{\Sigma(t)} \mathcal{D}(x,t)\, d\mu(x). \tag{3}$$

Here, μ specifies the substrate measure used to define $\mathcal{D}$. For discrete elements, the counting measure reduces this integral to a sum over the localized elements.

The architectural pillars need not occupy physically disjoint regions of the substrate. Because the six pillars represent distinct functional requirements, their total expenditure partitions exactly into their additive costs (section II E), yielding $Z_{\text{net}}(t) = \sum_i Z_i(t)$. This strict accounting ensures that even if a single physical mechanism multiplexes multiple pillars, its operational cost is uniquely apportioned without double-counting.

The *Operational Action* S_Φ of a structural history Γ over an observation interval $[0, T]$ is defined as its cumulative operational cost:

$$\begin{aligned} S_\Phi[\Gamma; T] &= \int_0^T Z_{\text{net}}(t; \Gamma)\, dt \\ &= \int_0^T \int_{\Sigma(t)} \mathcal{D}(x, t; \Gamma)\, d\mu(x)\, dt. \end{aligned} \tag{4}$$

Comparing competing systems requires evaluating this action over a common observation interval, a shared resource unit, and a consistent convention for identifying the system boundary.

The dimensions of S_Φ inherit the units of the operational toll c. If c represents dissipated energy per operation, S_Φ quantifies macroscopic dissipated energy. If c is defined as a dimensionless action per operation, S_Φ becomes a pure, dimensionless count of state transitions. In arbitrary physical systems, a dimensionless Operational Action can always be recovered by dividing S_Φ by a characteristic reference cost.

While the static architecture identifies the functions instantiated by the system, the Operational Action records

the exact thermodynamic burden of their realized operation. This cost accumulates along any history, whether or not the system ultimately persists. Relating this accumulated Operational Action to the probability of survival requires a specific failure model, formalized in section IV.

B. The Exponential Filter

If persistence is maintained by state transitions, and each transition risks scrambling information into the substrate noise floor, then a history of N transitions survives only if every step survives. Independent errors compound multiplicatively, yielding an exponential filter.

Consider a structural history Γ accumulating operational action $S_\Phi[\Gamma]$ (equation (4)). Because the substrate possesses a finite noise floor, each elementary transition carries an intrinsic risk of information loss. Let q be the probability that a single elementary transition maintains causal coherence.

In QSS, successive transitions are statistically independent dissipative events. The probability that Γ maintains coherence is therefore multiplicative:

$$P(\text{survival} \mid \Gamma) = q^{S_\Phi[\Gamma]} \tag{5}$$

Using the identity $q^C = e^{C \ln q}$ with $C = S_\Phi[\Gamma]$, and defining $\beta \equiv -\ln q > 0$ (positive since $0 < q < 1$), we have $S_\Phi[\Gamma] \ln q = -\beta S_\Phi[\Gamma]$, yielding the survival probability distribution:

$$P(\text{survival} \mid \Gamma) = e^{-\beta S_\Phi[\Gamma]} \tag{6}$$

Here $\beta \equiv -\ln q$ is the substrate's inverse operational tolerance, a geometric invariant of the channel. For discrete substrates, q is determined by the channel geometry; in continuous media, β emerges from the substrate's noise spectral density.

This is the informational analogue of the Boltzmann distribution $P \propto e^{-E/k_B T}$, with S_Φ as energy and β as inverse temperature. The Boltzmann form requires ensemble equilibrium and a thermal bath; the present form requires only independent transition risk along a single history and a substrate noise floor. The informational route recovers the same exponential law under strictly weaker assumptions: substrate geometry replaces thermodynamic equilibrium.

C. The Severity of the Exponential Filter

The operational severity of this filter can be quantified by comparing trajectories. Let $\Gamma_{\min}$ denote the minimal-action trajectory and Γ_{var} any non-minimal path with an action deviation $\delta S_\Phi > 0$. Over any observation window, the relative survival probability is:

$$\frac{P(\Gamma_{\text{var}})}{P(\Gamma_{\min})} = e^{-\beta \delta S_\Phi} \tag{7}$$

When the fundamental update cycle is much shorter than the observation timescale, even a small per-transition deviation compounds into a vast accumulated δS_Φ, and the survival ratio $P(\Gamma_{\text{var}})/P(\Gamma_{\min})$ falls to zero. The gap between the best trajectory and the next-best widens without bound.

Because survival probability depends exponentially on S_Φ, only histories approaching the minimum retain non-negligible observation probability:

> The **Selection Principle**: Among all configurations capable of encoding a given structural complexity, only those possessing the minimal possible Operational Action (S_Φ) asymptotically dominate the observable distribution.

Because S_Φ counts elementary transitions, it is the cumulative ledger of a system's dissipation history. The substrate does not optimize this ledger; it erases ledgers that fail to balance.

D. Eliminative vs. Teleological Selection

The severity of the exponential filter is a statement about the losers, not the winner. It says nothing about whether $\Gamma_{\min}$ is efficient, elegant, or well-suited to anything, only that every alternative to it decayed faster. A sub-optimal trajectory survives just as certainly as an optimal one, provided everything else was worse. What looks like design, foresight, or purpose is not a property of the winner; it is the absence of its competition.

E. Minimum Action as Survivorship

This eliminative mechanism is formalized as:

Proposition 1 (Minimum Action as Survivorship)**.** *In QSS, the observable trajectories of any persistent system are those achieving the minimal Operational Action S_Φ among all candidate trajectories.*

Proof. By equation (6), the survival probability of any history Γ is $P(\text{survival} \mid \Gamma) = e^{-\beta S_\Phi[\Gamma]}$. For any non-minimal trajectory Γ_{var} with $\delta S_\Phi = S_\Phi[\Gamma_{\text{var}}] - S_\Phi[\Gamma_{\min}] > 0$, equation (7) gives $P(\Gamma_{\text{var}})/P(\Gamma_{\min}) = e^{-\beta \delta S_\Phi}$. As δS_Φ accumulates over the observation window, this ratio vanishes. Therefore, only trajectories achieving the minimal Operational Action retain non-negligible observation probability. The observable set is the argmin of $S_\Phi[\Gamma]$. □

Standard quantum mechanics grounds the Principle of Least Action in phase interference. The Selection Principle offers a distinct, classical mechanism (informational

elimination) by which candidate physical histories are selected: the substrate itself acts as a filter, and what remains observable is only what has not yet dissipated.

V. OPEN SYSTEMS: NECESSITY WITHOUT ADDITIVITY

The Selection Principle and the Principle of Least Action were derived in the Quiescent Steady State: the zero-flux limit where no net energy or information enters or leaves. Many systems, however, exchange flux with their environment to persist. We show that the necessity of the six pillars survives this extension unchanged; only the net impedance (Z_{net}, section II E) requires re-derivation in flux-driven capacity.

A. Pointwise Necessity and Operational Latency

Because the substrate continuously filters for causal coherence, survival is an instantaneous requirement, not a time-averaged one.

Proposition 2 (Pointwise Necessity)**.** *For any system persisting past instant t, all six pillars must be structurally instantiated at t, regardless of external flux.*

Proof. By the counterfactual failure modes of section II C, the absence of any pillar drives the corresponding primitive to zero ($c \to 0$, $K \to 0$, or $f \to 0$), leaving the system entirely vulnerable to its corresponding failure mode. Under continuous environmental perturbation, the instantaneous hazard rate diverges, driving the probability of surviving any finite interval dt to zero. The requirement is instantaneous and flux-independent. □

Flux alters a pillar's *operation*, not its *existence*. During phases of rapid growth or unbounded resource influx, a control mechanism may exhibit low activation ($f_{\text{GOV}} \approx 0$), minimizing active coordination expenditure. Yet its *structural capacity* must exist at t; if the mechanism is fundamentally absent, the system fails to persist the moment it reaches environmental saturation.

B. Open Systems and the Dynamic Budget

Strict additivity ($\bar{Z}_{\text{net}} = \sum_i Z_i$) relies on the Quiescent Steady State, where orthogonal domains prevent interference. In open systems, external flux raises the dissipation ceiling; as operations scale against this moving ceiling, resource contention forces non-linear cross-coupling $\Delta(t)$ between previously orthogonal pillars:

$$Z_{\text{net}}(t) = \sum_i Z_i(t) + \Delta(t). \tag{8}$$

A stochastic derivation of $\Delta(t)$ from substrate-native properties is developed in appendix A; three regimes emerge as its limits, where ρ is the dimensionless coordination load:

1. **Linear-Response Regime (Low ρ):** $\Delta \to 0$; QSS additivity is the leading-order behavior.
2. **Saturation-Divergence Regime (High ρ):** Coordination cost diverges as the dissipation ceiling is approached, forcing structural pauses (apoptosis, circuit breakers) to shed load.
3. **Correlated-Demand Regime (Non-Markovian):** Relaxing the memoryless assumption recovers the standard heavy-traffic generalization, with the $\rho/(1-\rho)$ form as its memoryless special case.

The derivation does not alter the necessity claim. The six pillars remain structurally mandatory regardless of the specific form of $\Delta(t)$; the idealized limit where $\Delta(t) \to 0$ recovers QSS additivity.

That derivation also provides a unified account of these three regimes, which were previously analyzed independently. Linear-response recovery, heavy-traffic divergence, and the correlated-demand generalization were each discovered independently in unrelated fields. Here they emerge as three limits of one substrate function.

VI. UNIVERSALITY MAPPING: ARCHITECTURE AND DYNAMICS

The core mechanics of persistence are identifiable across systems of any substrate. Mapping both the static architecture (the six pillars) and the operational dynamics (the dissipation trade-off and selection filter) translates domain-specific phenomena into identical mechanical constraints. The following mappings demonstrate architectural universality across distinct physical and informational domains. Detailed operationalizations belong to applied persistence mechanics; here, we illustrate the underlying structural invariants.

A. Biological Cell

Formalizing the biological concept of autopoiesis [15]:

Architecture: Capacity as metabolic hardware; Map as genome; Protocol as genetic code and signaling machinery; Governor as apoptosis and cell-cycle checkpoints; Toll as the thermodynamic cost of proofreading; Margin as ATP reserves and stress buffers.

Dynamics: The cell's basal metabolic rate ($\mathcal{D}$) is the product of its ATP cost per molecular reaction (c), proteome complexity (K), and metabolic turnover rate (f). An organism cannot simultaneously maximize genomic

complexity and reproductive speed without exceeding the local thermodynamic ceiling. Trajectories that overdraw this budget undergo exponential suppression via starvation or competitive exclusion.

Cross-Domain Transfer: Rather than treating aging as a correlational pattern of stochastic damage, this architectural defines senescence as the progressive degradation of operational pillars across nested sub-systems, revealing structural failure modes analogous to resource exhaustion in computational systems.

B. Ecological Network

Formalizing ecological ascendency and network thermodynamics [16]:

Architecture: Capacity as biomass and solar energy capture; Map as trophic structure and niche topology; Protocol as nutrient cycling and metabolic exchange; Governor as carrying-capacity and predator-prey feedbacks; Toll as thermodynamic heat loss per trophic transfer; Margin as biodiversity and structural redundancy.

Dynamics: Total ecosystem respiration ($\mathcal{D}$) is the product of heat lost per transfer (c), food-web complexity (K), and the rate of biomass turnover (f). Because solar influx rigidly caps $\mathcal{D}$, overly complex food webs with inefficient trophic transfers are systematically pruned by famine and ecological collapse, leaving only the least-action trophic topology.

Cross-Domain Transfer: Mapping ecological networks to energy grids and financial markets reveals that cascading collapse obeys the same fixed dissipation ceiling, making grid load-shedding and market circuit-breakers candidate tools for anticipating tipping points in biodiversity loss.

C. Large Language Model

Formalizing a young domain still establishing its foundations:

Architecture: Capacity as inference compute and GPU memory bandwidth; Map as parametric weights and attention matrices; Protocol as tokenization, layer connections, and key-value (KV) caching; Governor as output constraints and behavioral guardrails; Toll as autoregressive latency and state-update friction; Margin as unused context window, processing redundancy, and decision margin.

Dynamics: The model's active dissipation burden ($\mathcal{D}$) is the product of compute friction per generated token (c), context and parameter footprint (K), and generation velocity (f). In autoregressive generation, attempts to maximize context length and reasoning depth without lowering parameter activation costs hit a hard memory bandwidth and compute ceiling. The Operational Action acts as a strict evolutionary filter on AI architectures: models that fail to minimize this product succumb to cumulative attention noise (hallucination) or catastrophic inference costs, driving the structural selection of minimal-action survivors (e.g., KV-cache distillation, sparse routing, or linear-time state-space models).

Cross-Domain Transfer: The comparatively young field of AI can draw on cross-disciplinary knowledge without re-deriving it: long-context degradation is currently benchmarked and treated empirically. If the Saturation-Divergence Regime correctly describes KV-cache saturation, its onset is a hazard-rate problem already solved by Weibull analysis.

Contention Regimes: The three regimes of appendix A are visible in inference. Short-context generation resembles the Linear-Response Regime. As context length approaches the memory-bandwidth ceiling, generation quality collapses in a manner resembling the Saturation-Divergence Regime, the KV-cache saturation and attention degradation already documented in long-context models. Non-uniform, correlated attention patterns (repetition, recency bias) degrade worse than a memoryless baseline would predict, suggesting the Correlated-Demand Regime as a candidate lens, not a demonstrated one.

D. Further Applications

The same six-pillar mapping extends to domains further removed from physical substrates. Two examples illustrate distinct payoffs of that extension, briefly rather than in full: one shows the mapping makes previously incomparable systems commensurable; the other shows the architecture binds even a domain with no physical substrate at all.

- **Institutional Organizations:** A shared dissipation metric shows how bureaucratic decay in a growing firm and signal decay in a fiber-optic network are directly comparable structural failures.
- **Scientific Enterprise:** Reframing Planck's observation that science advances one funeral at a time as the thermodynamic survival of funded believers over verification friction shows that theoretical paradigms are constrained by the same dissipation limits as cells or firms.

VII. CONCLUSION

Fluctuation theorems, master equations, and Langevin dynamics describe the statistics of transitions; the architecture of persistence describes the structure that must

exist before any such statistics can be defined. For the fields of adaptation and self-organization, this shifts the fundamental question: before a system can adapt or self-organize, it must possess a minimal architecture permitted to exist.

We have developed that architecture and a mechanism of selection under the stated dynamical assumptions. The six pillars organize the requirements of persistence; the Operational Action measures cumulative dissipation; and the Selection Principle describes the resulting survival bias when failure hazard is proportional to that action.

The shared architecture provides a starting point for transferring knowledge across disciplines. Because this universal mapping projects every persistent system onto the same six-pillar substrate, it functions as a Rosetta stone across disciplines. Hard-won solutions in one domain become candidates for transfer to another: protocol engineering that prevents signal decay in a fiber-optic network can suggest mechanisms for preventing fragmentation in a corporate hierarchy. In emerging domains such as AI, where the pillars are present but unnamed, this vocabulary potentially lets systems be engineered for resilience without re-deriving structural knowledge from scratch.

The same mapping enables direct quantitative comparison among competing systems in a shared environment. By evaluating how competing organisms, firms, or technical architectures manage their finite dissipation budgets, relative operational resilience can be assessed prior to empirical failure.

Organizational decline, ecosystem collapse, and the limits of biological longevity have been described through domain-specific frameworks. By defining exactly what is required to persist, this mechanics recasts them as structural failures under a single, unified mechanics: disparate phenomena become comparable instances of the same dissipation constraint, rather than loose analogies.

The Complexity Trade-off is a pre-observational constraint on structural complexity, coordination speed, and energetic efficiency. At a binding dissipation ceiling, increasing one factor requires a compensating reduction in the product of the other two; given the ceiling, any two positive factors bound the third.

The most consequential limiting case is fundamental physics: whether standard physical law is itself a substrate-specific instantiation of this same invariant. Further exploration will determine whether this architectural boundary holds universally. Candidate mathematical languages for testing this include Information Geometry (continuous substrates), Arithmetic Geometry (discrete substrates), and Sheaf Theory (compositional substrates).

Natural avenues for the extension include systems composed of nested persistent subsystems, what it takes for a system to arise from disorder, and a fuller account of decay itself: this paper identifies failure modes, without characterizing the trajectory a system follows on the way there, or whether decay has its own structure worth deriving.

Persistence Mechanics does not explain why structure arises. It explains why only certain architectures remain to be observed. Persistence requires no intent, no optimization, and no forward-looking agency, only that the substrate continuously filter non-viable configurations. What remains observable is the minimal architecture of persistence.

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Appendix A: An Example of the Dynamic Budget via Queueing-Theoretic Contention

Open systems violate the orthogonality that made $\bar{Z}_{\text{net}}$ strictly additive in QSS. To quantify the resulting contention between Protocol and Governor, we derive the cross-coupling $\Delta(t)$ in equation (8) from discrete substrate properties.

1. Coordination capacity and load

Protocol and Governor share the substrate that performs all state transitions. The Margolus-Levitin limit bounds the minimum time between distinguishable operations, yielding a substrate-level coordination rate $f_{\max}$. In QSS, the two pillars operate on orthogonal domains (their demands share no microstate), so their loads add. With $f_{\text{PRO}} + f_{\text{GOV}} \ll f_{\max}$, cycles remain effectively empty and orthogonality holds ($\Delta = 0$). The dimensionless coordination load is therefore

$$\rho(t) = \frac{f_{\text{PRO}}(t) + f_{\text{GOV}}(t)}{f_{\max}} \tag{A1}$$

2. Temporal independence and the geometric waiting distribution

The load above governs instantaneous rates, not waiting-time statistics. We model retry behavior as a memoryless process: each cycle presents pending events symmetrically, with service probability $1 - \rho$ per event. This is a modeling hypothesis about temporal independence, not a consequence of orthogonality. The waiting time follows a geometric distribution with mean

$$\langle k \rangle = \sum_{k=0}^{\infty} k\, \rho^k (1 - \rho) = \frac{\rho}{1 - \rho}. \tag{A2}$$

3. The coordination cross-term

Each pending cycle incurs the fixed toll c_0 (the dimensionless Landauer scale; on thermal substrates this projects to $k_B T \ln 2$). The total waiting cost for both streams is

$$\Delta_{\text{total}}(t) = \frac{c_0\, f_{\max}\, \rho}{1 - \rho}. \tag{A3}$$

The QSS baseline (equation (1)) already prices the linear coordination costs $Z_{\text{PRO}} + Z_{\text{GOV}} \approx c_0(f_{\text{PRO}} + f_{\text{GOV}})$ at $\rho \ll 1$. The cross-coupling $\Delta(t)$ is the excess beyond this baseline:

$$\Delta(t) = \Delta_{\text{total}}(t) - c_0\, f_{\max}\, \rho = \frac{c_0\, f_{\max}\, \rho^2}{1 - \rho}. \tag{A4}$$

This recovers the three regimes of section V B: linear-response recovery as $\rho \to 0$, saturation divergence as $\rho \to 1$, and correlated-demand behavior under heavy-tailed waiting distributions.

4. Scope and validity

This construction assumes discrete cycles (the Margolus-Levitin bound) and symmetric, memoryless retry. Strongly correlated, heavy-tailed, or long-range-dependent arrivals fall outside its scope.